# SMART INDIA HACKATHON 202

SMART INDIA HACKATHON 2026

**Problem Statement ID – 26189**

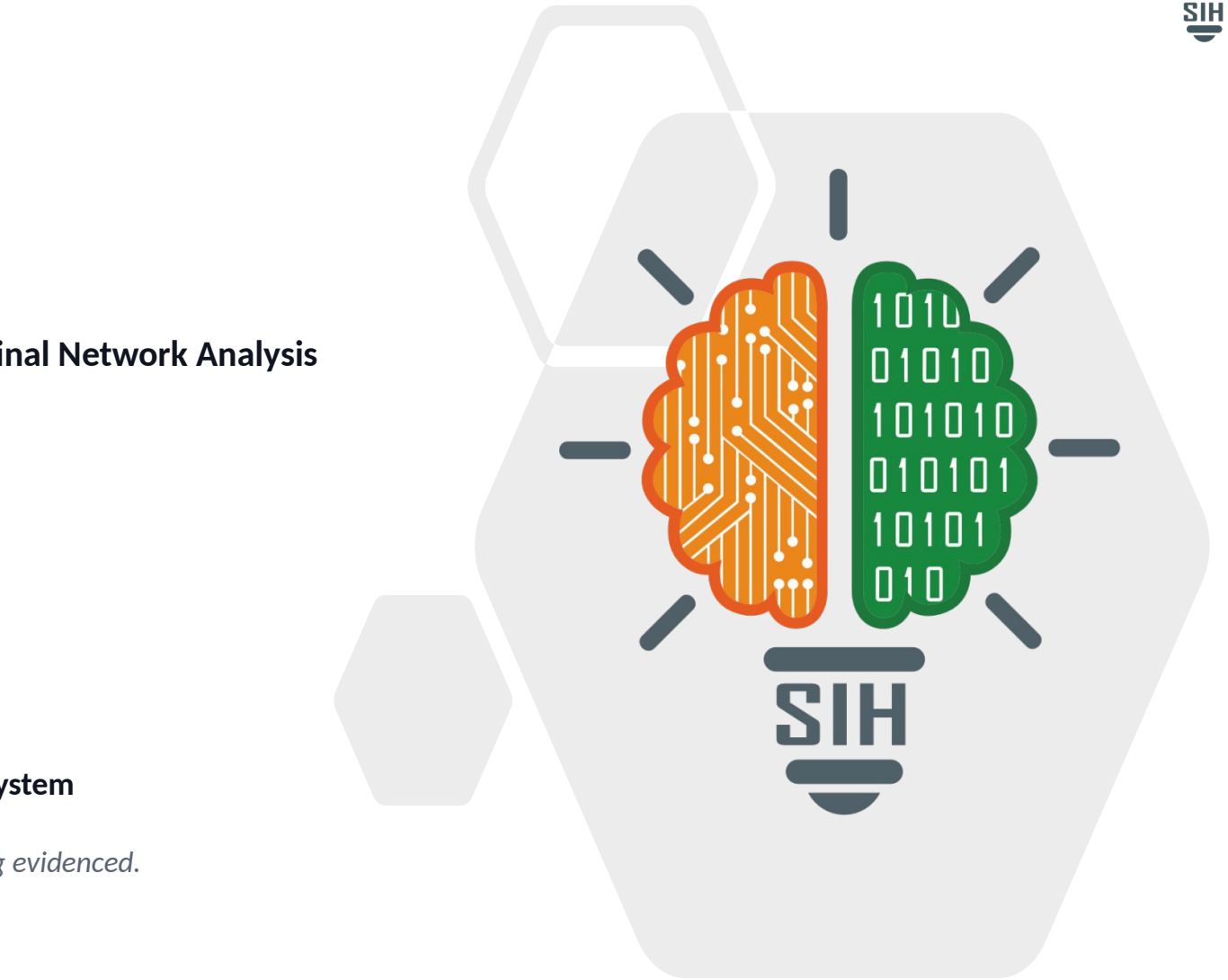
**Problem Statement Title – AI-Powered Criminal Network Analysis System**

Theme – Blockchain & Cybersecurity

PS Category – Software

Team ID – <Team ID>

Team Name – Team Innovexa




# DRISTI

**Evidence-Centric Criminal Network Intelligence System**

दृष्टि · "sight, perception"

*One case graph. Many analytical lenses. Every finding evidenced.*

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# IDEA AND PROPOSED SOLUTION

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## THE PROBLEM

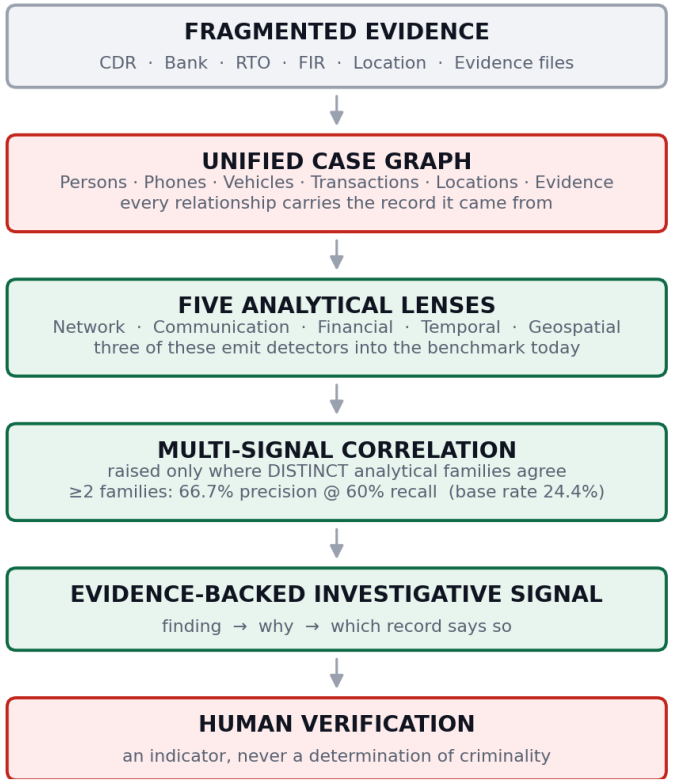
- Investigation data is fragmented across CDR, banking, RTO, FIRs, locations and evidence files.
- Relationships between people, phones, vehicles, transactions and places are hard to establish across systems that do not talk to each other.
- Officers spend days cross-referencing records and reconstructing timelines by hand.
- A detected relationship is only useful if it can be traced back to the record that evidences it.
- Any single analytical signal, on its own, produces more noise than leads.

### THE GAP IS NOT DISCOVERY

An investigator often already suspects the connection. What takes weeks is assembling it from disconnected systems — and then showing, for each link, which record it came from. A tool that surfaces names without that trail adds work instead of removing it.

## PROPOSED SOLUTION

DRISTI builds one case-level investigation graph that unifies heterogeneous evidence and lets multiple analytical lenses examine the same network.




## CORE INNOVATION

### 01 PROVENANCE-AWARE EVIDENCE GRAPH

Evidence is part of the data model, not a reporting feature. Every relationship and every finding can be walked back to the source record that produced it.

### 02 MULTI-SIGNAL CORRELATION ACROSS ANALYTICAL FAMILIES

Signals from distinct analytical families are correlated to strengthen investigative leads; single-detector outputs remain leads rather than corroborated findings. Model output is never counted as a corroborating family.

### 03 GROUND-TRUTH EVALUATION FRAMEWORK

A labelled synthetic investigation with known relationships and planted patterns, so the system can be measured rather than demonstrated — including where it fails.

**Recorded fact ≠ derived inference ≠ model prediction**

*DRISTI stores which kind of claim it is making.*

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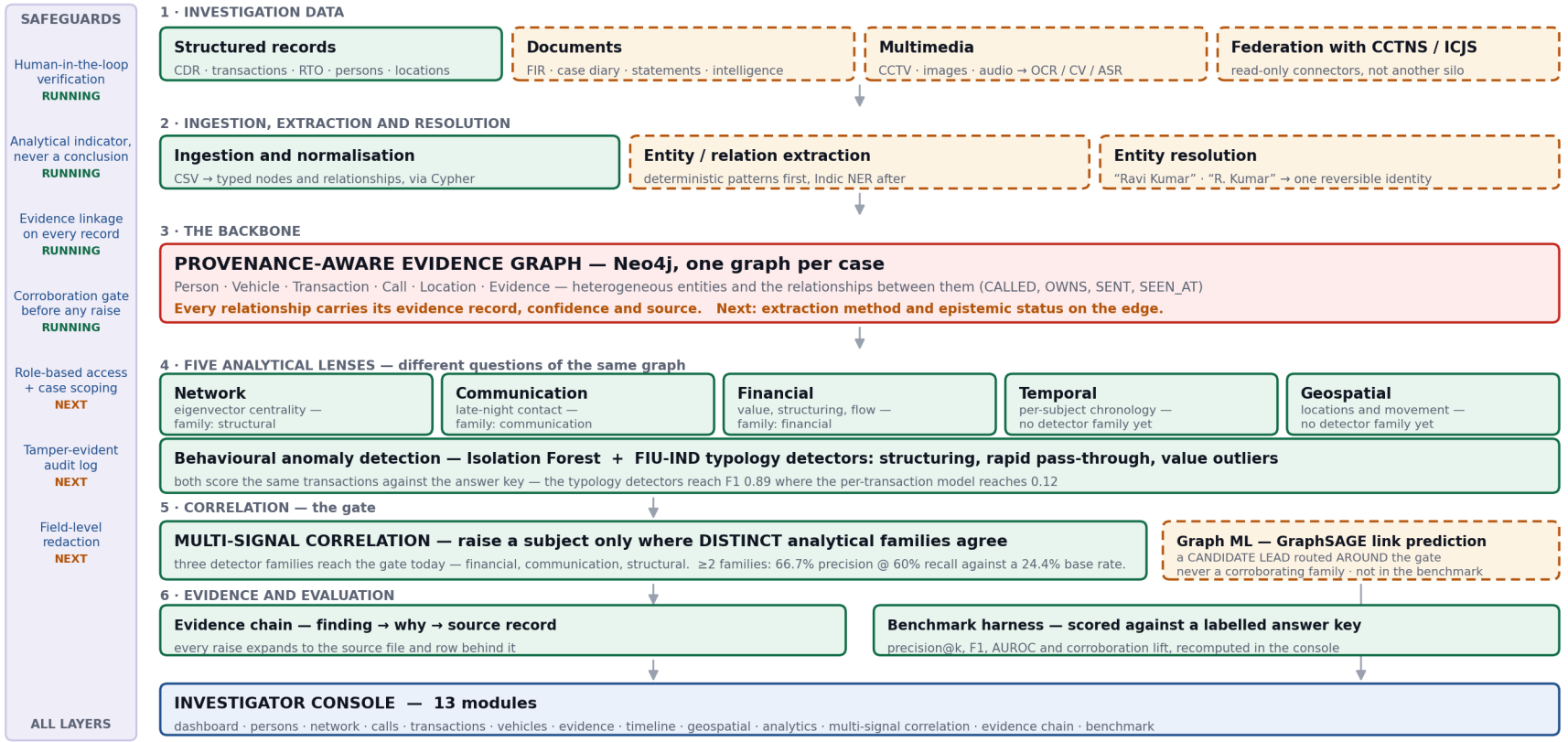
# TECHNICAL APPROACH

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## DRISTI — system architecture

One case-level knowledge graph in Neo4j, read by separate analytical modules — every finding tied back to its source record, every raise requiring corroboration, an investigator in front of the output.

running in the console today · designed, next phase · the backbone




*Green blocks run in the console today and can be opened on request; amber dashed blocks are designed and scheduled. Five analytical lenses exist in the console; three of them — financial, communication, structural — emit the detector families that feed the correlation gate measured here. Graph ML sits on its own branch by design: a link prediction is a candidate lead, never a corroborating family, and it does not enter the benchmark.*

### TECHNOLOGY STACK

- **Data and backend**
  Python · pandas · Neo4j · Cypher · Streamlit
- **Graph analytics**
  NetworkX · Neo4j GDS · Plotly
- **Machine learning**
  scikit-learn · Isolation Forest
- **Financial intelligence**
  FIU-IND typology detectors — structuring, rapid pass-through, value outliers
- **Evaluation**
  labelled synthetic corpus · benchmark harness in the console
- **NLP and entity resolution**
  IndicNER · MuRIL · phonetic blocking
- **Graph ML**
  GraphSAGE link prediction — candidate leads only
- **Security**
  RBAC · case scoping · tamper-evident audit log

● *running today* ○ *next phase*

*No GPU and no outbound network call — deployable inside an existing police network.*

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# FEASIBILITY AND VIABILITY

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## A WORKING PROTOTYPE, NOT A CONCEPT

Live Neo4j case graph · 13 modules · 82 subjects · 2,522 calls · 254 transactions · 2,811 evidence records




| | person | level | score | families | family_list | detectors | n_dete |
|---|---|---|---|---|---|---|---|
| 0 | Girish Kumar | HIGH | 0.9666 | 3 | communication, financial, structural | late_night_contact, network_centrality, rapid_passthrough | |
| 1 | Tarun Yadav | MEDIUM | 0.9759 | 2 | financial, structural | network_centrality, rapid_passthrough, structuring | |
| 2 | Harish Naidu | MEDIUM | 0.9744 | 2 | financial, structural | network_centrality, structuring | |

*Network Graph and Multi-Signal Correlation, live. HIGH 1 · MEDIUM 17 · LOW 27. HIGH requires all three currently implemented detector families to agree; the thresholds are fixed, not tuned to the answer key.*

### CURRENTLY DEMONSTRATED

Neo4j case graph · network visualisation · entity traversal · call, transaction, timeline and geospatial analytics · Evidence Desk · anomaly detection · typology detectors · multi-signal correlation · evidence chain · benchmark evaluation.

## CHALLENGES AND MITIGATION

| Challenge | How DRISTI addresses it |
|---|---|
| **Fragmented identities** | Conservative entity resolution — merge only on a shared identifier or a corroborating attribute, and every merge reversible by the officer. |
| **False positives from any one analytic** | Correlation across distinct analytical families. On our labelled corpus, ≥2 families gives 66.7% precision at 60% recall against a 24.4% base rate. |
| **A score mistaken for a finding of guilt** | Indicators labelled as indicators on every screen, a human-verification step in front of every output, predictions stored apart from recorded facts. |
| **Sensitive CDR, FIR and financial data** | Role-based access with case-level scoping, field-level redaction and a tamper-evident audit log — access bound to assignment, every read accountable. |
| **Messy, unstructured source documents** | Staged extraction: deterministic patterns for numbers and registration marks, a locality gazetteer for places, an Indic NER model for names last. |
| **Standardized benchmarks are limited across the literature reviewed** | We built a labelled corpus and harness of our own. It gives controlled prototype evidence, not a prediction of field performance; next step is the same harness on real annotated records with NCRB. |
| **Graph growth across a large case** | Indexed Cypher queries and modular analytics — each lens reads the graph independently, so the console scales by module, not by rewrite. |

*Feasible on open-source components, ordinary hardware and a modular architecture — no GPU, no cloud dependency, no outbound traffic from a police network.*

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# IMPACT AND BENEFITS

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## FOR THE INVESTIGATING OFFICER

- One unified case picture instead of manually cross-referencing disconnected sources — the assembly work that currently takes days collapses into a single load.
- Relationship discovery, timeline reconstruction and communication, financial and spatial pattern analysis against the same graph.
- Immediate evidence traceability: every subject screen answers “what else is this person connected to?” and “which record says so?” side by side.

## FOR NCRB AND THE INSTITUTION

- An evidence-backed analytical workflow — the chain from an insight to the originating record is part of the data model, so every finding is auditable and traceable rather than asserted.
- A standardised case-level investigation graph, with auditable access.
- Designed to federate with CCTNS and ICJS through read-only connectors rather than becoming another parallel silo.

## FOR THE CITIZEN — WHY THE SAFEGUARDS ARE THE FEATURE

A system that names people from inferred patterns can do real harm to someone innocent. DRISTI makes no autonomous criminal conclusions: an indicator is never presented as a determination, a person surfaced by the system can always be traced back to the specific records that surfaced them, and access to a citizen's call and financial records is bound to a case assignment and logged. Proportionality and purpose limitation are design constraints here, not a compliance annexe.

**No autonomous criminal conclusions · human verification in front of every output · evidence traceability on every record · case-bound access · auditable use of sensitive data.**

## ANOMALY ≠ CRIMINALITY

*AI assists the investigator. It does not replace investigative judgement.*

## MEASURED, NOT ASSERTED

Work in this field is validated almost entirely by case study. We built a labelled synthetic investigation — 82 subjects, 2,522 calls, 254 transactions, 20 syndicate members, planted patterns — and scored our own analytics against the answer key.

**Key individuals by network connectivity** P@5 1.00 · P@10 1.00 · P@20 1.00

**Evidence linkage** 100% of records resolve to a source evidence record

**Transaction anomaly** generic outlier model F1 0.12 / AUROC 0.51 · **structuring typology detector F1 0.89**

*That gap is the finding, not a defect: structuring is built to look ordinary one transaction at a time, so no per-row model can see it. Measuring it is how we knew to add typology detectors rather than a bigger model.*

## MULTI-SIGNAL CORRELATION — LABELLED SYNTHETIC CORPUS

Gate: a subject is raised only on ≥2 distinct analytical families (financial · communication · structural). Thresholds fixed, not tuned to the answer key.

**66.7% precision**

**at 60% recall · base rate 24.4%**

Evaluated population 82 persons
Ground-truth positives 20
**Flagged at ≥2 families 18**
**TP 12 · FP 6 · FN 8 · TN 56**

*Measured on our labelled synthetic investigation corpus. Controlled prototype evidence that the pipeline recovers a known structure and that design choices can be compared under identical conditions — not a prediction of field performance.*

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# RESEARCH AND REFERENCES

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## RESEARCH FOUNDATIONS


**Criminal network analysis**

Sparrow, M. — "The application of network analysis to criminal intelligence: an assessment of the prospects", Social Networks 13(3), 1991.

Morselli, C. — "Inside Criminal Networks", Springer, 2009. Brokerage, and why covert networks are deliberately sparse.

**Network analytics**

Bonacich, P. — "Power and Centrality: A Family of Measures", American Journal of Sociology 92(5), 1987.

Blondel, V. et al. — "Fast unfolding of communities in large networks", J. Stat. Mech., 2008. arxiv.org/abs/0803.0476

**Anomaly detection**

Liu, F.T., Ting, K.M., Zhou, Z-H. — "Isolation Forest", IEEE ICDM, 2008.

**Graph ML and entity resolution**

Hamilton, W., Ying, R., Leskovec, J. — "Inductive Representation Learning on Large Graphs" (GraphSAGE), NeurIPS 2017. arxiv.org/abs/1706.02216

Christen, P. — "Data Matching", Springer, 2012. Blocking, pairwise scoring and transitive closure.


## TECHNOLOGY, DOMAIN AND LEGAL GROUNDING


**Systems this federates with**

National Crime Records Bureau — CCTNS and the Inter-operable Criminal Justice System (ICJS). ncrb.gov.in

Neo4j Graph Data Science — centrality, community detection and pathfinding at scale. neo4j.com/docs/graph-data-science

**Financial intelligence**

Financial Intelligence Unit – India — cash transaction reporting thresholds and layering typologies; the basis for the financial indicators. fiuindia.gov.in

**Language resources**

AI4Bharat — IndicNER and MuRIL, for names in Indic scripts and transliterated forms in FIR text. ai4bharat.org

**Legal and privacy framework**

Bharatiya Nagarik Suraksha Sanhita, 2023 — production of documents and search; the evidence chain is built to survive this scrutiny.

Digital Personal Data Protection Act, 2023 — purpose limitation and accountability; why access is bound to case assignment and logged.

K.S. Puttaswamy v. Union of India, (2017) 10 SCC 1 — necessity and proportionality; why the system refuses to escalate on a single unverified signal.


## OUR CONTRIBUTION

A provenance-aware case graph, a correlation layer across distinct analytical families, and a labelled ground-truth evaluation framework for evidence-centric criminal-network investigation. Across the literature we reviewed, existing evaluations are often case-study based, with limited standardization across datasets and metrics, which makes systems hard to compare directly. We intend to release our corpus and harness, including the negative results they produce — the per-transaction outlier model at F1 0.12 among them.